

10.2 Biased data ↕

- Watch: Telling the truth with data visualisation (part 2) [12:27 mins]

Watch the following video, which describes three common types of bias and explains the importance of adjusting for inflation when time series data represents long time periods.

This is useful as it is important to ensure that the data used to create data visualisations is representative of appropriate samples and accurate values.



[Transcript](#)

<https://learn.usc.edu.au/courses/35242/files/3138524?wrap=1>

Source: © University of the Sunshine Coast (2023).

Download the following slides for note taking: [ICT702 Telling the truth with data visualisation \(part 2\)](https://learn.usc.edu.au/courses/35242/files/3168490?wrap=1).<https://learn.usc.edu.au/courses/35242/files/3168490?wrap=1>

Bias in data occurs when the data sample does not represent the population under study. The more common causes of biased data are defined below.

Selection bias

Selection bias is when the sample of the population from which the data has been drawn has not been appropriately randomised prior to data collection. Therefore, it does not truly represent the intended population. It can occur for several reasons. For example, selecting a particular mode of communication, such as mobile phone rather than a landline, may preclude a certain demographic from participating in a study.

Simpson's paradox

Simpson's paradox refers to an effect that occurs when the trend identified in subsets of data is reversed when the subsets are aggregated into one whole dataset. The issue is a form of selection bias caused by choosing a sample that does not represent the whole population.

Survivor bias

Survivor bias also results from sample data not being a true reflection of the population under investigation. In survival bias, the sample data has a disproportional number of observations with positive outcomes for a specific event; in other words, selection of the sample has been impacted by a positive response to a prior situation.

□ Watch: Biased and unbiased samples [7.55 mins]

This video is useful as it describes what a sample is and explains the difference between a biased sample and an unbiased sample. Understanding why biased samples must be avoided is crucial for telling the true data story.

As you watch, consider:

- How might biased data impact organisational decision making?
- Why is it important to avoid the use of biased data?



Source: © Scam Squad Math (2021).

□ Watch: Sources of bias [6.59 mins]

This video is useful as it explains about artefacts, which are artificial patterns caused by deficiencies in the data collection process. When analysing data, it is extremely important to distinguish between facts (aspects of the real relationship) and artefacts gathered during data collection.

As you watch, consider:

- How could a lack of understanding about the data collection/selection/classification process influence analysis and the final data story?
- Why should this be important to the data story audience?



Source: © Wild About Statistics (2015).

□ Activity: Cognitive biases – three common types illustrated

Reflect on this webpage about three common types of cognitive bias. Note the webpage was published in April 2022.

As you read take time to consider and make notes on:

- how each visualisation represents the type of bias being explained
- what techniques you would use to illustrate each type of bias, and why you would make that selection.

Ensure that you can justify your thinking and bring your answer to the workshop.

Hay, J. & Robinson, J. (2022, April 16). [Cognitive biases: Three common types illustrated](https://www.visualcapitalist.com/cp/cognitive-biases-three-common-types-illustrated/)  [\(https://www.visualcapitalist.com/cp/cognitive-biases-three-common-types-illustrated/\)](https://www.visualcapitalist.com/cp/cognitive-biases-three-common-types-illustrated/). *Visual Capitalist*.